



DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

--

CLASS

--

INDEX
NUMBER

--

PRELIMINARY EXAMINATION 2025 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

4045/02

Paper 2

5 August 2025

2 hours

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Section A

Answer **all** questions.

Section B

Answer **one** question.

The number of marks is given in brackets [] at the end of each question or part question.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total number of marks for this paper is 70.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator or 3.142.

Total Marks

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f} \right)^2}$$

Section A (62 marks)

Answer **all** the questions in this section.

1 (a) Calculate $\sqrt[4]{(-9)^2 + \frac{7}{8}}$.

Answer [1]

(b) $\frac{\pi}{2}$ $-\left(\frac{5}{4}\right)^2$ 120% $\sqrt{2}$

Write these numbers in order, starting with the largest.

Answer , , , [2]
largest

2 (a) Given that $\frac{a^8 \times a^2}{a^{-4}} = a^n$, find n .

Answer $n =$ [2]

(b) Given that $\sqrt{27} \div 9^2 = 3^m$, find m .

Answer $m =$ [2]

- 3 The table shows the results from a survey of the number of siblings that each student from a particular class has.

Number of siblings	0	1	2	3	4	5
Frequency	6	9	13	p	3	2

- (a) Explain why it is not possible to find a value of p such that the median number of siblings is 1.5.

Answer

 [1]

- (b) If the mean is 1.95 siblings, show that $p = 7$.

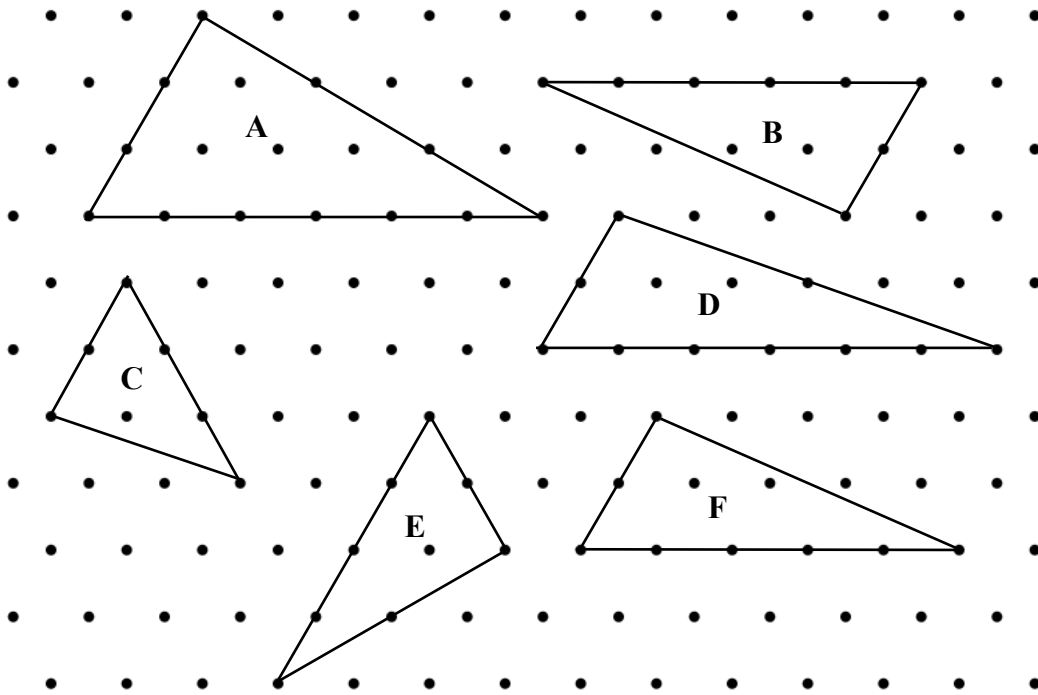
Answer

[2]

- (c) A student from the class is selected at random.
 Find the probability that this student has at least 3 siblings.

Answer [1]

- 4 (a) The diagram shows 6 triangles drawn on a 1 cm triangular grid. The triangles are labelled A, B, C, D, E and F.



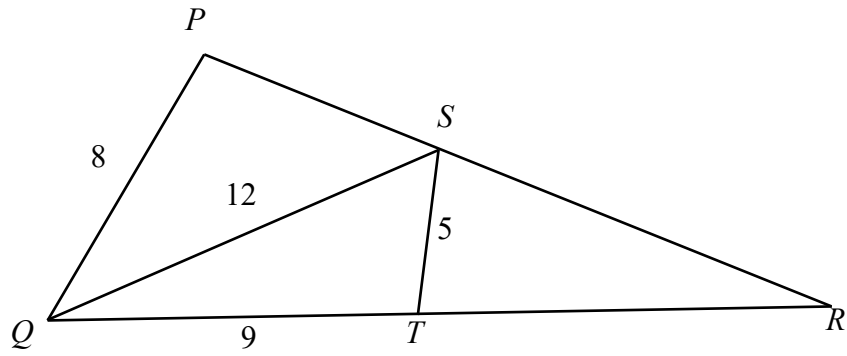
- (i) Write down two triangles that are congruent to each other.

Answer Triangle and triangle [1]

- (ii) Name a triangle that is similar to triangle A.

Answer Triangle [1]

(b)



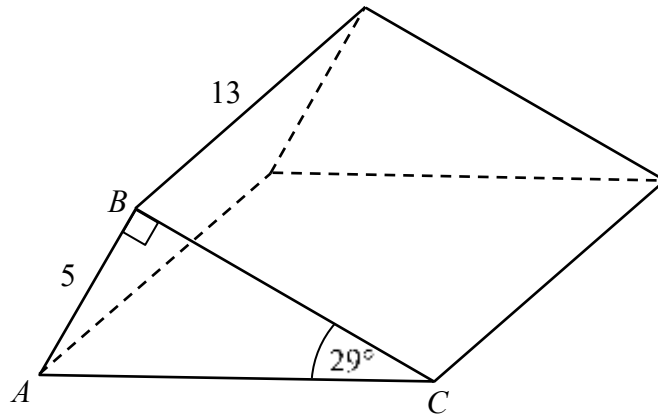
Triangle PQR is similar to triangle TSQ .

$PQ = 8$ cm, $QS = 12$ cm, $QT = 9$ cm and $ST = 5$ cm.

Calculate the length of TR .

Answer cm [3]

5



The diagram shows a prism. Three of its faces are rectangles.

Triangle ABC is the cross-section of the prism.

$AB = 5$ cm, angle $ABC = 90^\circ$ and angle $ACB = 29^\circ$.

The length of the prism is 13 cm.

- (a) Show that $AC = 10.313$ cm, correct to 3 decimal places.

Answer

[2]

- (b) Find the **total** surface area of the prism.

Answer cm^2 [4]

- 6 (a) Rearrange this formula to make w the subject.

$$w - 4 = \frac{2w + 3}{x}$$

Answer $w = \dots\dots\dots$ [3]

- (b) Shan thinks of a prime number p .
 She subtracts 6 from p and multiplies the result by p .
 She then adds 12 to the answer.
 Her final answer is twice of p .

- (i) Write down an equation in p and show that it simplifies to

$$p^2 - 8p + 12 = 0.$$

Answer

[2]

- (ii) Using factorisation, solve $p^2 - 8p + 12 = 0$ to find the prime number p .

Answer [2]

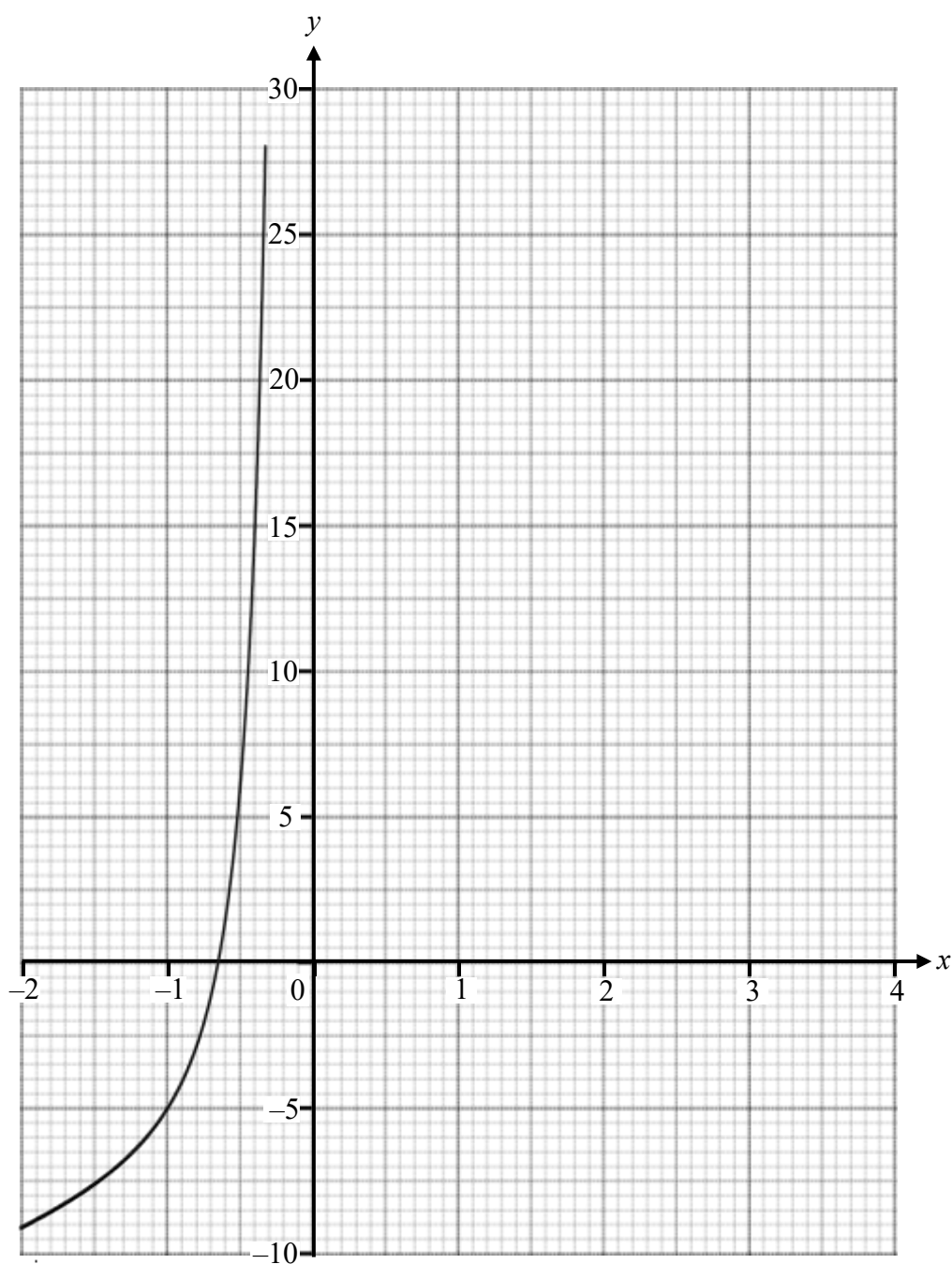
- 7 (a) Complete the table of values for $y = \frac{3}{x^2} + 2x - 6$.

x	0.3	0.4	0.5	0.6	1	2	3	4
y	27.9	13.6	7	3.5	-1	-1.3	0.3	

[1]

- (b) On the grid, plot and draw the graph of $y = \frac{3}{x^2} + 2x - 6$ for $0.3 \leq x \leq 4$.

The graph of $y = \frac{3}{x^2} + 2x - 6$ for $-2 \leq x \leq -0.3$ has been drawn for you.



[2]

- (c) Explain why the graph does not intersect the y -axis.

Answer
 [1]

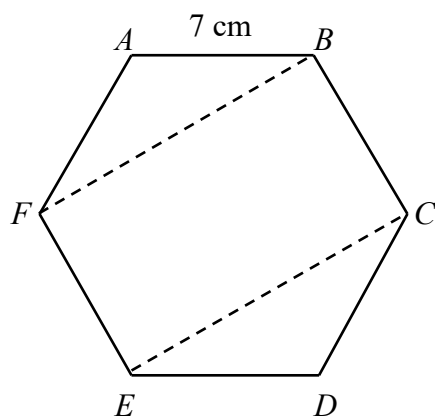
- (d) By drawing a suitable tangent, find the gradient of the curve when $x = -1$.

Answer [2]

- (e) Use your graph to find the values of x when $\frac{3}{x^2} + 2x - 6 = 0$.

Answer [3]

- 8 The diagram shows a regular hexagon $ABCDEF$ with side length of 7 cm.



- (a) Show that angle $ABF = 30^\circ$.

Answer

[2]

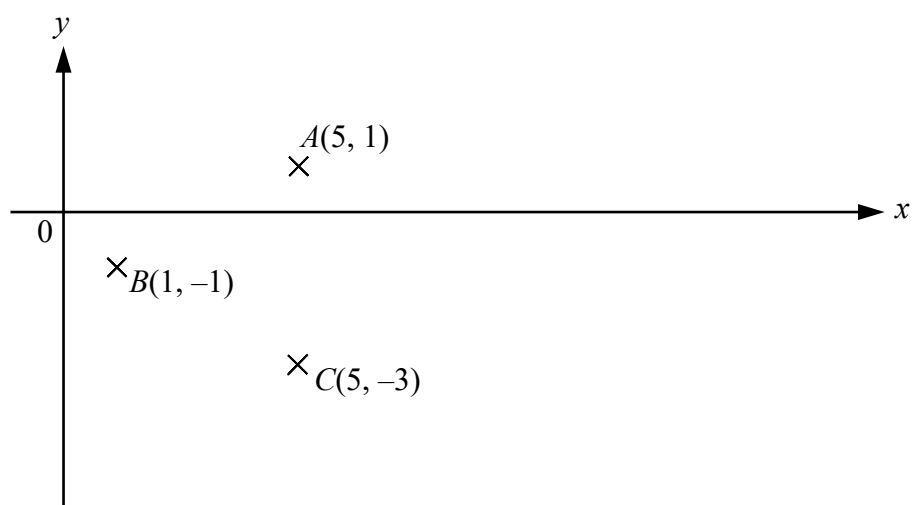
- (b) Explain why the quadrilateral $BFEC$ is a rectangle.

Answer
.....
.....
.....
..... [2]

- (c) Find the area of rectangle $BFEC$.

Answer cm^2 [3]

- 9 A is the point $(5, 1)$, B is the point $(1, -1)$ and C is the point $(5, -3)$.



- (a) Find the length of AB .

Answer units [2]

- (b) Find the equation of BC .

Answer [3]

(c) $ABCD$ is a kite with $BD = 10$ units.

(i) Write down the coordinates of D .

Answer (.....,) [1]

(ii) Write down the equation of BD .

Answer [1]

(iii) Find the area of the kite $ABCD$.

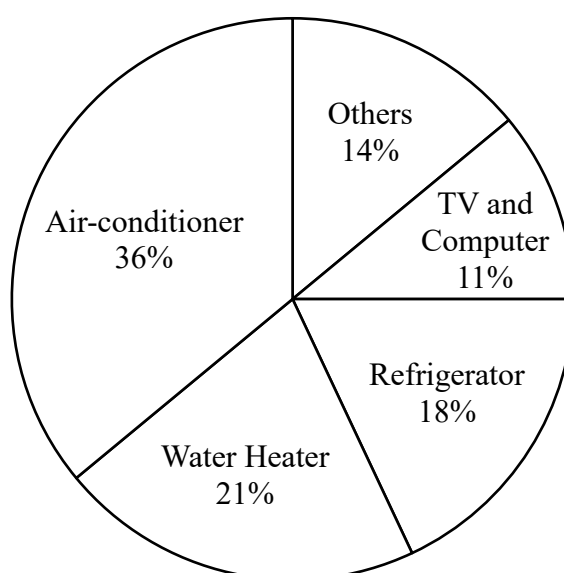
Answer units² [2]

- 10 Below is some information about the household electricity consumption in Singapore.

**Average household electricity consumption per month (in kilowatt hours)
categorised by housing type for the year 2024**

Housing Type	Average electricity consumption per month (in kWh)
1-room / 2-room flats	175
3-room flats	275.1
4-room flats	379
5-room flats / Executive Apartments	461.4
Private Apartments and Condominiums	519.5
Landed Properties	1203.8

**Percentage breakdown of electricity use for each type of electrical appliance in a
typical Singapore household**



- (a) Estimate the amount of electricity used by the refrigerator in a typical 3-room household in Singapore in a month.

Answer kWh [1]

In the year 2024, it costs 29.67 cents on average for the use of 1 kWh of household electricity, excluding 9% Goods-and-Services Tax.

- (b) Estimate the cost of electricity use by a typical 4-room household in the year 2024, including 9% Goods-and-Services Tax. Give your answer correct to the nearest dollar.

Answer \$ [3]

- (c) Mr Tan is looking to reduce his family's electricity costs.
They live in a 5-room flat and use an average of 500 kWh of electricity per month.
The percentage of electricity they use for air-conditioner units is the same as the percentage for a typical Singapore household.

- (i) Estimate the amount of electricity used by the air-conditioner units in Mr Tan's household in a month.

Answer kWh [1]

- (ii) Currently, Mr Tan and his family use the air-conditioner units for an average of 8 hours each day. Mr Tan claims that if they switch on the air-conditioner for 5 hours each day instead, they can get their electricity usage below the 2024 average household electricity use per month for 5-room flats.

Is Mr Tan correct? Explain your answer.

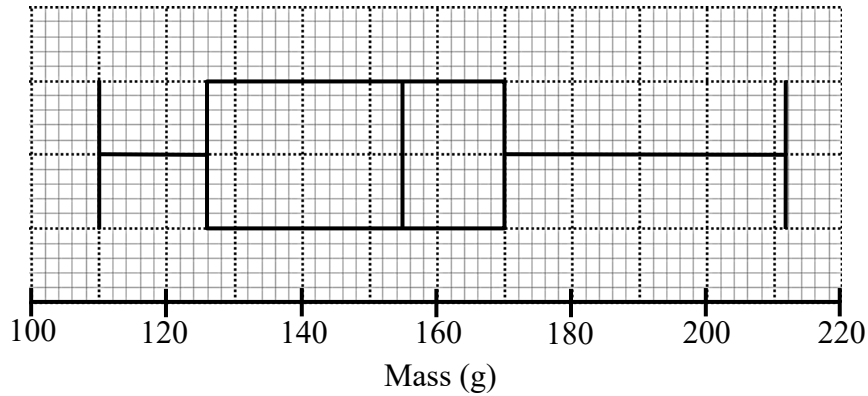
Answer

[3]

Section B (8 marks)

Answer **one** question from this section. Each question carries 8 marks.

- 11 (a)** The box-and-whiskers plot summarises the masses of apples produced from farm A.



- (i)** Find the median mass.

Answer g [1]

- (ii)** Find the range of the masses.

Answer g [1]

- (iii)** Find the interquartile range of the masses.

Answer g [1]

- (iv)** In farm B, the median mass of apples is 144 g, the range of the masses is 92 g and the interquartile range is 50 g.

Hui says the apples produced from farm B have more consistent masses.

Is her statement correct? Give a reason for your answer.

Answer

 [1]

- (b) The table summarises the heights of 50 apple trees in farm A.

Height, h m	$1 \leq h \leq 2$	$2 < h \leq 3$	$3 < h \leq 4$	$4 < h \leq 5$	$5 < h \leq 6$
Frequency	2	5	11	19	13

- (i) Yen says the range of heights of these 50 trees is 5 metres.
Explain why she could be wrong.

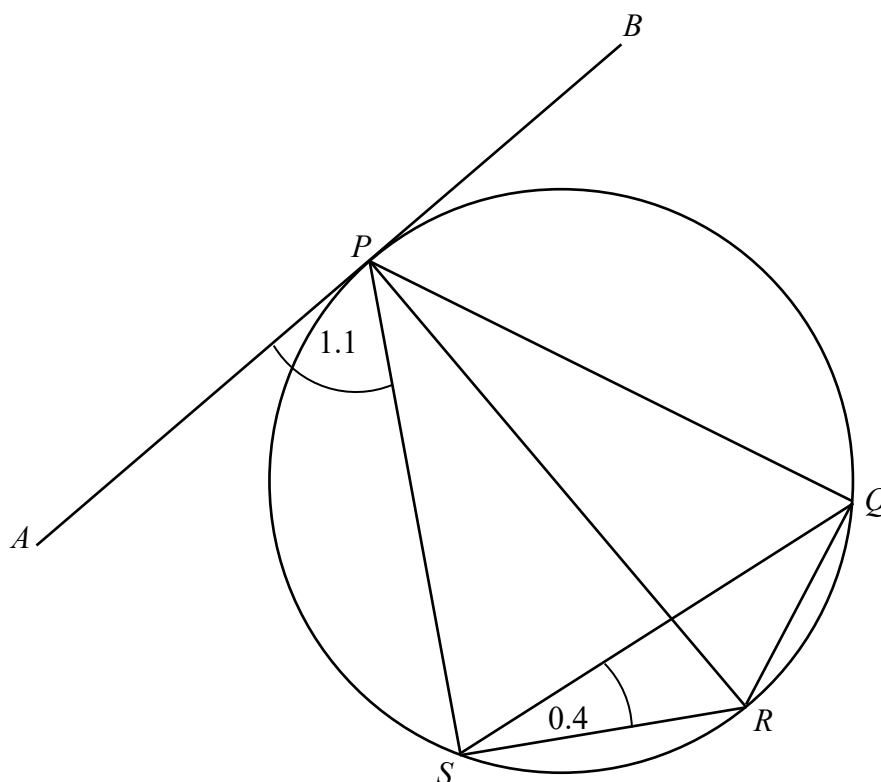
Answer

 [1]

- (ii) Two of the trees are picked at random.
Find the probability that one tree is at most 4 metres in height and the other tree is more than 5 metres in height.

Answer [3]

12



$PQRS$ is a circle with diameter PR .

AB is a tangent to the circle at P .

Angle $QSR = 0.4$ radians and angle $APS = 1.1$ radians.

- (a) (i) Find angle RPQ in radians and give a reason for your answer.

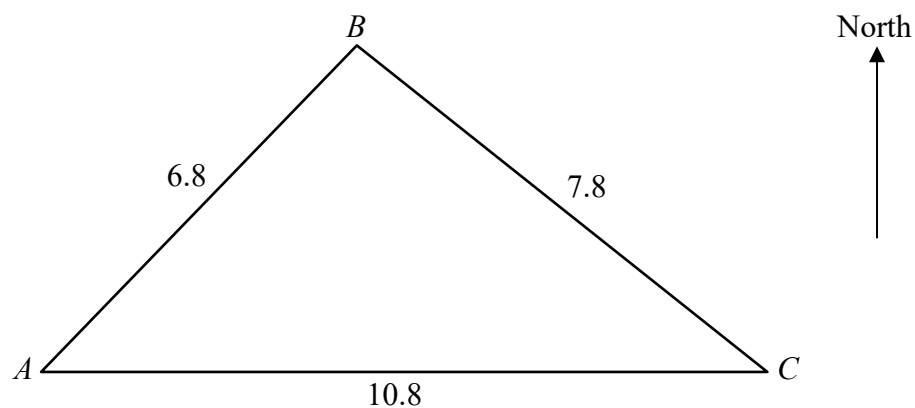
Angle $RPQ = \dots\dots\dots$ Reason $\dots\dots\dots$

$\dots\dots\dots$ [1]

- (ii) Find angle SPR in radians.

Answer $\dots\dots\dots$ [1]

(b)



The diagram shows a triangular field, ABC , on horizontal ground.

C is due east of A .

$AB = 6.8$ m, $BC = 7.8$ m and $AC = 10.8$ m.

- (i) Find the bearing of B from A .

Answer $^{\circ}$ [4]

- (ii) At B , there is a tree of height 4.6 metres.
Calculate the angle of elevation of the top of the tree from C .

Answer $^{\circ}$ [2]

END OF PAPER

